

Lesson 50 Practice: Data Representation, Number Systems & Logic

JKSSB Computer Awareness

October 10, 2026

Provenance: 16 original JKSSB-pattern questions and five paraphrased adaptations of official JKSSB PYQs: Computer Instructor/Operator 2025 Q1–Q4 and Website Operator 2026 Q73. Adaptations are not verbatim reproductions.

Q1. *[PYQ-adapted: JKSSB Computer Instructor/Operator 2025 Q1; official paper; paraphrased]*
What is the decimal ASCII code for the dollar sign (\$)?

- A. 32
- B. 30
- C. 34
- D. 36

Q2. *[PYQ-adapted: JKSSB Computer Instructor/Operator 2025 Q2; official paper; paraphrased]*
Convert the binary number 11010101.01_2 to decimal.

- A. 200.25_{10}
- B. 198.25_{10}
- C. 213.25_{10}
- D. 209.25_{10}

Q3. *[PYQ-adapted: JKSSB Computer Instructor/Operator 2025 Q3; official paper; paraphrased]*
Which statement about UTF-8 is correct?

- A. It is fixed-length and uses exactly 8 bits for every character.
- B. It is variable-length and compatible with ASCII in the ASCII range.
- C. It assigns Unicode code points only to Latin letters.
- D. It stores every character using 8 bytes.

Q4. *[PYQ-adapted: JKSSB Computer Instructor/Operator 2025 Q4; official paper; paraphrased]*
What is the binary representation of 278_{10} ?

- A. 100010110_2
- B. 101010110_2
- C. 100110110_2
- D. 110010110_2

Q5. *[PYQ-adapted: JKSSB Website Operator 2026 Q73; official provisional key; paraphrased]*
Which binary numeral has the value 10_{10} ?

- A. 1010_2
- B. 1001_2
- C. 1100_2
- D. 1110_2

Q6. [Original] What is the base of the octal number system?

- A. 2
- B. 8
- C. 10
- D. 16

Q7. [Original] Which digit is not permitted in an octal numeral?

- A. 0
- B. 5
- C. 7
- D. 8

Q8. [Original] What is the decimal value of $2F_{16}$?

- A. 31
- B. 32
- C. 47
- D. 63

Q9. [Original] Convert 1111111_2 to octal.

- A. 177_8
- B. 177_2
- C. 77_8
- D. 177_{16}

Q10. [Original] In BCD, how is the decimal number 25 represented?

- A. 11001_2
- B. 0010 0101
- C. 0101 0010
- D. 11101_2

Q11. [Original] Which statement is **NOT** correct?

- A. Original ASCII uses 7-bit codes.
- B. Unicode assigns code points across many writing systems.
- C. UTF-8 is variable-length.
- D. BCD and Unicode are interchangeable names for character encoding.

- Q12.** *[Original]* Which option describes the relationship between Unicode and UTF-8 correctly?
- A. Unicode assigns code points; UTF-8 encodes them using a variable-length byte sequence.
 - B. UTF-8 is another name for BCD.
 - C. Unicode is a 4-bit code for each decimal digit.
 - D. UTF-8 is always fixed at 8 bytes per character.
- Q13.** *[Original]* In common 24-bit RGB, the three 8-bit channels represent:
- A. Red, green, and blue
 - B. Red, grey, and black
 - C. Row, grid, and bit
 - D. Read, generate, and store
- Q14.** *[Original]* For inputs $A = 1$ and $B = 0$, what is the output of XOR?
- A. 0
 - B. 1
 - C. 2
 - D. It depends on a clock
- Q15.** *[Original]* Which gate outputs 1 only when both inputs are 1?
- A. OR
 - B. XOR
 - C. AND
 - D. NOT
- Q16.** *[Original]* For $A = 1$ and $B = 1$, the NAND output is:
- A. 0
 - B. 1
 - C. 2
 - D. Undefined
- Q17.** *[Original]* A NOT gate has:
- A. One input and the opposite output
 - B. Two inputs and output 1 only when both are 1
 - C. Two inputs and output 1 when they differ
 - D. Three inputs representing RGB
- Q18.** *[Original]* Evaluate the statements about a common 24-bit RGB representation:
- I. It uses red, green, and blue channels.
 - II. Three 8-bit channels total 24 bits per pixel.
 - III. Every image format must use exactly 24-bit RGB.

- A. I only
- B. I and II only
- C. II and III only
- D. I, II, and III

Q19. *[Original]* What is 1011_2 in decimal?

- A. 9
- B. 10
- C. 11
- D. 13

Q20. *[Original]* When converting a binary numeral to hexadecimal by grouping bits, how many bits are placed in each complete group?

- A. 2
- B. 3
- C. 4
- D. 8

Q21. *[Original]* For inputs $A = 0$ and $B = 0$, what are the outputs of AND, OR, XOR, and NAND, in that order?

- A. 0, 0, 0, 1
- B. 1, 0, 0, 0
- C. 0, 1, 1, 1
- D. 1, 1, 0, 1

Q22. *[JKSSB FAA-pattern]* The binary number 1010 equals which decimal value?

- A. 10
- B. 8
- C. 12
- D. 101

Q23. *[JKPSI-pattern]* How many distinct values can 3 bits represent?

- A. 8
- B. 3
- C. 6
- D. 9

Q24. *[SSC CGL-pattern]* Which logic gate gives output 1 only when all inputs are 1?

- A. AND
- B. OR

- C. NOT for two inputs
- D. Printer gate

Q25. *[JKSSB-pattern]* (MSQ: select ALL correct options.) Which binary-decimal conversions are correct?

- A. $101 = 5$
- B. $1000 = 8$
- C. $111 = 7$
- D. $101 = 6$

Q26. *[SSC CGL-pattern]* (MSQ: select ALL correct options.) Which statements about logic gates are correct?

- A. AND outputs 1 only if all inputs are 1.
- B. OR outputs 1 if any input is 1.
- C. NOT inverts its single input.
- D. AND outputs 1 when all inputs are 0.

Q27. *[Original]* What is the decimal value of binary 1010?

- A. 8
- B. 10
- C. 12
- D. 14

Q28. *[Original]* In hexadecimal notation, the digit A represents decimal:

- A. 8
- B. 9
- C. 10
- D. 11

Q29. *[Original]* BCD represents each decimal digit using:

- A. Four bits
- B. One byte
- C. Three bits
- D. Sixteen bits

Q30. *[Original]* Which logic gate outputs 1 only when its two inputs differ?

- A. XOR
- B. AND
- C. OR
- D. NOT

Q31. *[Original, MSQ drill]* (**MSQ: select ALL correct options.**) Which statements are correct?

- A. One hexadecimal digit represents four binary bits.
- B. A byte contains eight bits.
- C. BCD encodes each decimal digit in four bits.
- D. Decimal 10 is binary 1110.

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